

Program 9

Program-9/

```
|— namespaces.yaml
|— pods.yaml
```

Step 1: namespaces.yaml

```
apiVersion: v1 kind:
Namespace
metadata: name:
dev
---
```

```
apiVersion: v1 kind:
Namespace
metadata: name:
prod
```

Step 2: pods.yaml

```
apiVersion: v1 kind: Pod
metadata:
  name: frontend-pod
  namespace: dev labels:

  app: frontend
  env: dev

spec: containers:

  - name: frontend
    image: nginx ports:
    - containerPort: 80
---
```

```
apiVersion: v1
kind: Pod
metadata:

  name: backend-pod
  namespace: dev labels:
```

app: backend
env: dev

spec: containers:

- name: backend

image: nginx ports:

- containerPort: 80

apiVersion: v1

kind: Pod

metadata:

name: frontend-pod

namespace: prod

labels:

app: frontend

env: prod

spec:

containers:

- name: frontend

image: nginx ports:

- containerPort: 80

apiVersion: v1

kind: Pod

metadata:

name: backend-pod

namespace: prod

labels:

app: backend

env: prod

spec:

containers:

- name: backend

image: nginx ports:

- containerPort: 80

Step 3: Apply YAML files

```
kubectl apply -f namespaces.yaml
```

```
kubectl apply -f pods.yaml
```

step 4: Verify Pods by Namespace

```
kubectl get pods -n dev
```

```
kubectl get pods -n prod
```

Step 5: Filter Pods by Label

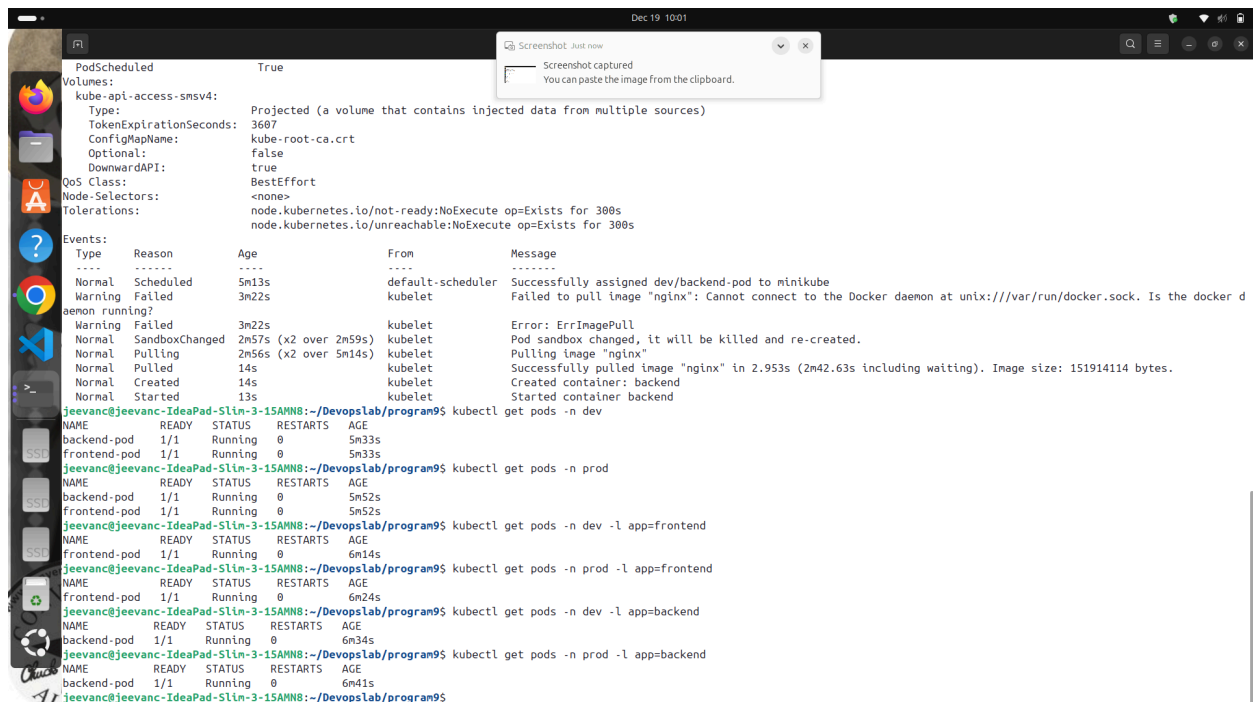
List frontend pods in dev

```
kubectl get pods -n dev -l app=frontend
```

List backend pods in prod

```
kubectl get pods -n prod -l app=backend
```

OUTPUT :



```
PodScheduled: True
Volumes:
  kube-api-access-smv4:
    Type: TokenExpirationSeconds: 3607
    ConfigMapName: kube-root-ca.crt
    Optional: false
    DownwardAPI: true
    QoS Class: BestEffort
    Node-Selectors: <none>
    Tolerations: node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                  node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type      Reason      Age      From      Message
  ----      -
  Normal    Scheduled   5m13s    default-scheduler    Successfully assigned dev/backend-pod to minikube
  Warning   Failed      3m22s    kubelet    Failed to pull image "nginx": Cannot connect to the Docker daemon at unix:///var/run/docker.sock. Is the docker d
  Warning   Failed      3m22s    kubelet    Error: ErrImagePull
  Normal    SandboxChanged 2m57s (x2 over 2m59s) kubelet    Pod sandbox changed, it will be killed and re-created.
  Normal    Pulling      2m56s (x2 over 5m14s) kubelet    Pulling image "nginx"
  Normal    Pulled       14s      kubelet    Successfully pulled image "nginx" in 2.953s (2m42.63s including waiting). Image size: 151914114 bytes.
  Normal    Created      14s      kubelet    Created container: backend
  Normal    Started      13s      kubelet    Started container backend
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/Devopslab/program9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           5m33s
frontend-pod  1/1     Running   0           5m33s
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/Devopslab/program9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           5m52s
frontend-pod  1/1     Running   0           5m52s
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/Devopslab/program9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           6m14s
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/Devopslab/program9$ kubectl get pods -n prod -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           6m24s
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/Devopslab/program9$ kubectl get pods -n dev -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           6m34s
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/Devopslab/program9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           6m41s
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/Devopslab/program9$
```